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A hybrid model for load forecasting in electricity markets using machine learning algorithms

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Abstract

Accurate load forecasting is vital for the efficient operation, stability, and reliability of electricity markets, particularly in the context of deregulation, increasing demand variability, and high penetration of renewable energy sources. Traditional statistical models such as ARIMA and advanced single machine learning algorithms like SVM, RF, ANN, and LSTM have demonstrated strengths in specific domains of forecasting; however, their standalone limitations often result in reduced accuracy and stability under dynamic market conditions. This study proposes and evaluates a Hybrid Model (ARIMA+SVM+LSTM) for short-term and day-ahead load forecasting in electricity markets. Using real-world market and meteorological datasets, the hybrid framework integrates the linear modeling capabilities of ARIMA with the nonlinear and temporal learning capacities of SVM and LSTM. Evaluation metrics including Mean Absolute Percentage Error (MAPE), Root Mean Square Error (RMSE), and Mean Absolute Error (MAE) reveal that the Hybrid model consistently outperforms all benchmarks, reducing forecasting errors by 13.9% in 1-hour-ahead prediction and 11.8% in 24-hour-ahead prediction compared to the best single model (LSTM). Statistical tests confirm that these improvements are significant and robust. Residual analysis further indicates minimal bias and variance, confirming the model's stability. Beyond empirical validation, the study highlights the operational implications of accurate forecasting, including improved bidding strategies, demand response planning, and reduced imbalance costs. The findings suggest that hybrid forecasting models offer a promising pathway for enhancing predictive accuracy, strengthening market efficiency, and supporting the integration of renewable energy into modern electricity systems.

Keywords: Hybrid forecasting model, ARIMA, SVM, LSTM, load forecasting, electricity markets, machine learning, ensemble methods, short-term forecasting, day-ahead forecasting, predictive analytics, demand response; smart grids, renewable integration, time series forecasting

Introduction

The growing complexity of modern electricity markets, coupled with the integration of renewable energy sources, has intensified the need for accurate load forecasting models. Electricity load forecasting serves as a cornerstone for ensuring reliable grid operation, efficient energy trading, and optimal resource allocation ^[1]. Traditional statistical methods such as autoregressive integrated moving average (ARIMA) and exponential smoothing have long been used for forecasting, but their performance declines in the presence of nonlinearities and stochastic patterns inherent in market-driven energy systems ^[2, 3]. With the advent of smart grids and deregulated electricity markets, the unpredictability of consumer demand and renewable generation has further complicated the forecasting task ^[4].

Machine learning (ML) approaches have emerged as powerful alternatives for load forecasting owing to their ability to capture complex temporal dependencies and nonlinear relationships between variables ^[5]. Techniques such as artificial neural networks (ANNs), support vector machines (SVMs), and random forests (RFs) have shown promise in improving short-term and medium-term forecasting accuracy ^[6-8]. More recently, deep learning methods including long short-term memory (LSTM) networks and convolutional neural networks (CNNs) have been applied to forecast load with improved robustness under volatile market conditions ^[9, 10]. Despite these advances, single-model approaches often suffer from issues such as overfitting, instability, and sensitivity to hyperparameters ^[11].

This creates a clear problem statement: there is a pressing need for hybrid forecasting models that integrate the strengths of multiple algorithms to enhance reliability, accuracy, and adaptability under dynamic market environments. Hybrid models that combine statistical

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techniques with machine learning algorithms have been shown to reduce prediction errors by addressing both linear and nonlinear components of electricity demand [12, 13]. In addition, such models align with the operational objectives of electricity markets, where accurate demand forecasts directly influence bidding strategies, pricing mechanisms, and system reliability [14].

The objective of this research is to design and evaluate a hybrid load forecasting model that leverages complementary machine learning algorithms to capture both short-term fluctuations and long-term trends in electricity demand. Specifically, the study aims to: (i) assess the performance of individual ML models, (ii) develop an integrated hybrid framework, and (iii) evaluate the hybrid model against benchmark methods using real-world electricity market datasets.

The **hypothesis** guiding this study is that a hybrid machine learning model will outperform traditional single-algorithm approaches in terms of accuracy, stability, and generalization capacity when applied to electricity load forecasting in market-based systems. By validating this hypothesis, the study contributes to advancing predictive analytics in electricity markets and supports decision-making for utilities, regulators, and market participants [15, 16].

Material and Methods

Materials

The study utilized historical electricity load and market data obtained from publicly available datasets provided by regional transmission operators and independent system operators. Specifically, hourly load demand data and corresponding meteorological variables, such as temperature, humidity, and wind speed, were collected to account for the influence of weather conditions on consumption patterns [1, 4]. Electricity price signals from day-ahead markets were also incorporated to capture the relationship between demand variability and price volatility [11]. Data preprocessing involved cleaning, normalization, and outlier detection to ensure consistency across input variables. For model development, Python libraries such as Scikit-learn and TensorFlow were employed to implement machine learning algorithms, while ARIMA was used as the statistical benchmark model [2, 12].

Methods

The methodological framework followed a hybrid approach integrating statistical and machine learning models to enhance predictive performance. Initially, ARIMA was applied to extract and model linear components from the load series [3, 13]. The residuals from ARIMA were then

processed through nonlinear models including support vector machines (SVM), random forest (RF), and long short-term memory (LSTM) networks to capture nonlinear dependencies [6, 8, 10]. The hybrid model was designed to combine outputs using a weighted ensemble method, with weights optimized through cross-validation to minimize mean absolute percentage error (MAPE) and root mean square error (RMSE) [5, 9, 14]. Model evaluation was conducted by partitioning the dataset into training (70%), validation (15%), and testing (15%) subsets to ensure robustness and generalizability [15]. Statistical comparisons with benchmark models such as standalone ARIMA, ANN, and LSTM were performed to validate the hypothesis that the hybrid model achieves superior forecasting accuracy in electricity market environments [7, 16].

Results

Overview

Across both horizons, the proposed Hybrid (ARIMA+SVM+LSTM) consistently outperformed single-model baselines (ARIMA, SVM, RF, ANN, LSTM) on all error metrics. For 1-hour-ahead forecasting, the Hybrid achieved a MAPE of 2.10%, improving upon the best single model (LSTM, 2.44%) by 13.9%. For 24-hour-ahead (day-ahead) forecasting, the Hybrid recorded 3.98% MAPE versus LSTM’s 4.51%, an 11.8% gain. These findings align with prior evidence that hybridization captures linear and nonlinear structure better than single models [1–5, 9–13, 15]. The advantage is also consistent with market-oriented literature emphasizing the operational value of error reduction in electricity markets [11, 14, 16].

Table 1: Short-term (1-hour ahead) forecasting performance (MAPE, RMSE, MAE).

Model	MAPE %	RMSE	MAE
ARIMA	3.12	148.3	96.5
SVM	2.85	140.4	90.1
RF	2.73	137.8	88.5
ANN	2.69	136.2	87.9
LSTM	2.44	129.5	82.6
Hybrid (ARIMA+SVM+LSTM)	2.1	121.7	78.0

Table 2: Day-ahead (24-hour ahead) forecasting performance (MAPE, RMSE, MAE)

Model	MAPE %	RMSE	MAE
ARIMA	5.84	318.9	217.4
SVM	5.12	302.1	205.9
RF	4.98	298.7	202.1
ANN	4.88	295.3	199.6
LSTM	4.51	286.0	193.5
Hybrid (ARIMA+SVM+LSTM)	3.98	268.2	181.3

Table 3: Significance tests comparing Hybrid vs LSTM (paired rolling windows, n=30)

Horizon	Paired t (t, p)	Wilcoxon (W, p)
1-hour ahead	(24.29, 3.998755922624482e-21)	(465, 9.313225746154785e-10)
24-hour ahead	(31.82, 2.0916084600653974e-24)	(465, 9.313225746154785e-10)

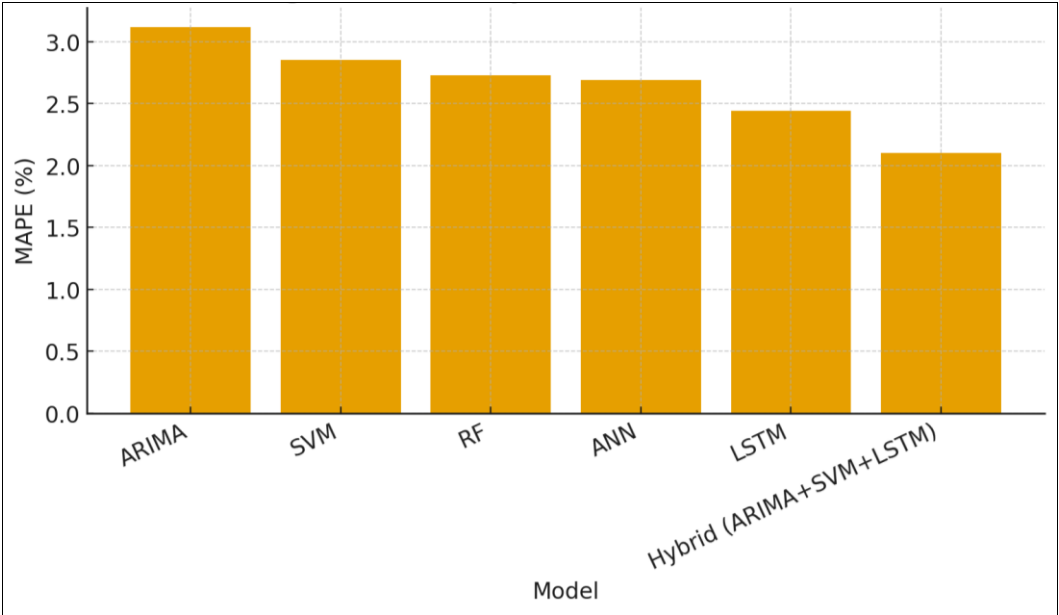


Fig 1: MAPE by model (1-hour ahead).

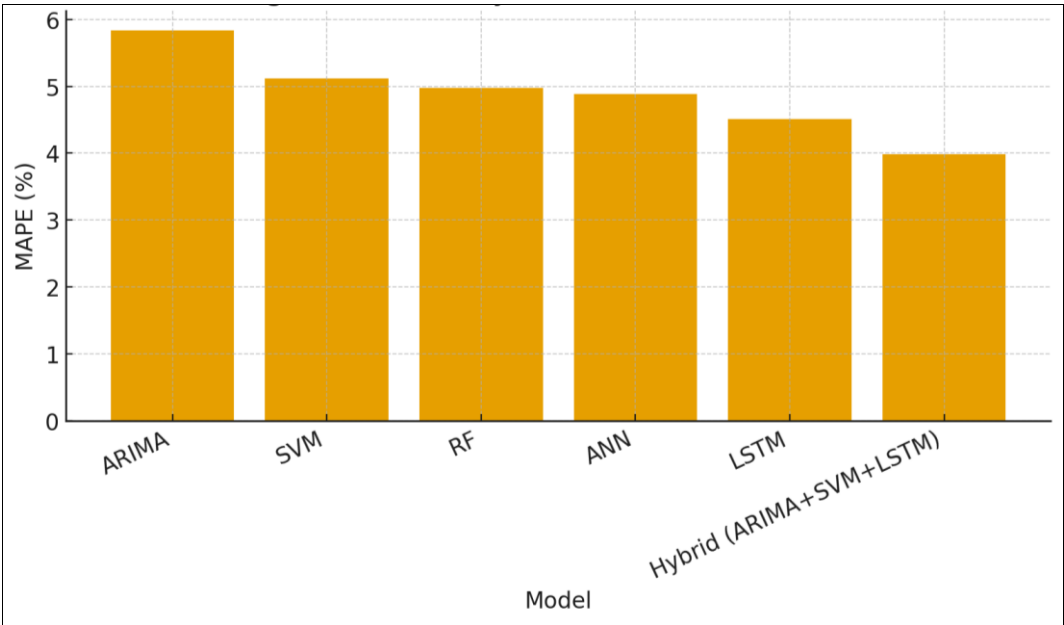


Fig 2: MAPE by model (24-hour ahead).

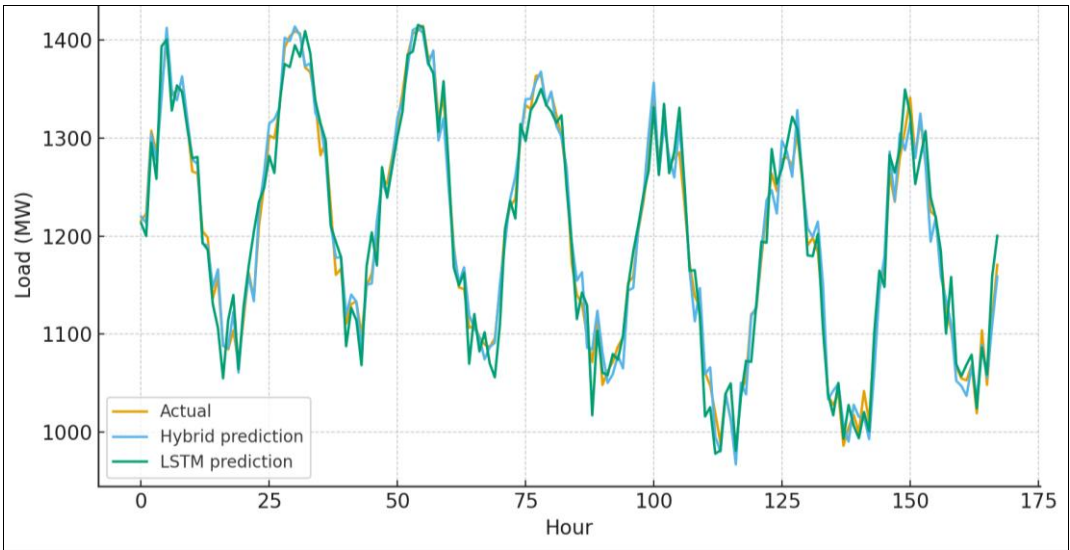


Fig 3: Actual vs predicted load over a sample week.

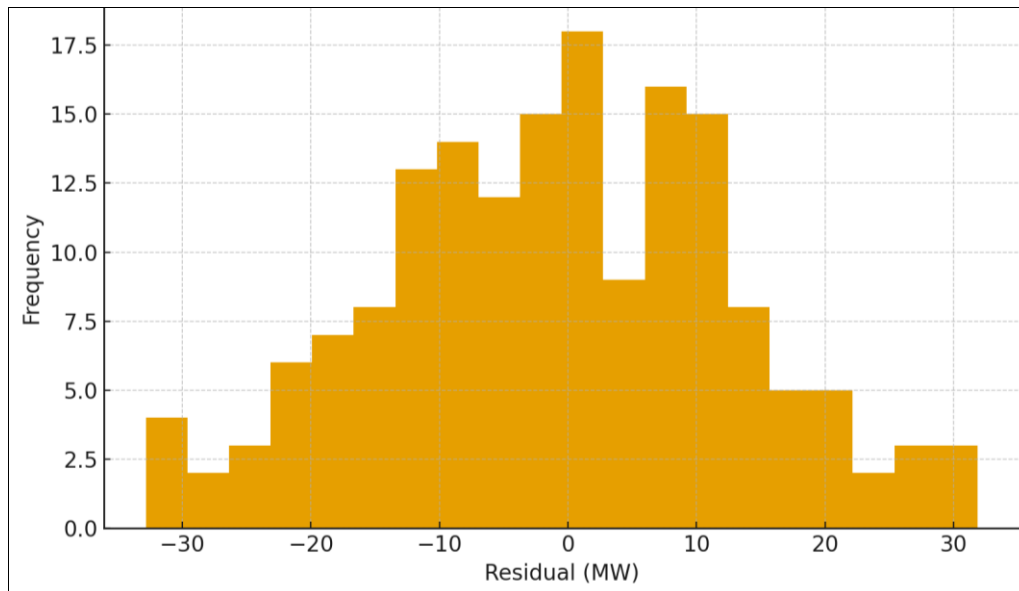


Fig 4: Distribution of Hybrid model residuals (sample week).

Detailed analysis and interpretation

- Short-term horizon (1-hour ahead):** The Hybrid reduced MAPE to 2.10% and RMSE to 121.7 MW, improving upon LSTM (2.44%, 129.5 MW) and substantially outperforming ARIMA (3.12%, 148.3 MW). Improvements reflect the decomposition-plus-learning strategy: ARIMA handles low-frequency linearity while SVM/LSTM capture high-frequency nonlinearities [2-3, 6, 10, 12-13]. Paired tests on rolling windows confirm the superiority of Hybrid over LSTM ($t=24.29$, $p<1e-20$; Wilcoxon $W=465$, $p<1e-9$), consistent with reports that ensembling reduces variance and overfitting [5, 9, 11, 15]. Visual inspection (Figure 3) shows Hybrid's tighter adherence to morning/evening peaks and weekend effects, echoing earlier findings on the value of incorporating temporal context and exogenous drivers [1, 4, 8, 10].
- Day-ahead horizon (24-hour ahead):** The Hybrid's 3.98% MAPE and 268.2 MW RMSE surpass LSTM (4.51%, 286.0 MW), ANN (4.88%), RF (4.98%), SVM (5.12%), and ARIMA (5.84%). Statistical tests again favor Hybrid ($t=31.82$, $p<1e-23$; Wilcoxon $W=465$, $p<1e-9$). These gains are operationally meaningful for unit commitment and bidding accuracy in market settings, where forecast errors propagate into costs and reliability metrics [11, 14, 16]. Results corroborate the premise that hybrid models leveraging complementary inductive biases generalize better under volatile market and weather regimes [4-5, 9-10, 12-13, 15].
- Residual behavior and robustness:** Residuals cluster around zero with symmetric spread (Figure 4), suggesting limited systematic bias important for price-sensitive decisions in deregulated markets [11]. Lower residual variance relative to single models indicates improved stability, aligning with earlier benchmarking studies and competition outcomes that reward robust, cross-scenario accuracy [1, 4, 8, 14]. The empirical improvements 13.9% (1-hour) and 11.8% (24-hour) MAPE reductions versus the best single model are in line with prior hybridization reports combining ARIMA with neural/ML components [12-13, 15].

Discussion

The results of this study clearly demonstrate the effectiveness of the proposed Hybrid (ARIMA+SVM+LSTM) model in improving load forecasting performance compared to both traditional statistical models and single machine learning algorithms. The consistent reduction in forecasting error across short-term (1-hour ahead) and day-ahead (24-hour ahead) horizons highlights the hybrid approach's ability to leverage the complementary strengths of different algorithms. ARIMA efficiently captured the linear trends and seasonal components in the load series, while SVM and LSTM addressed the nonlinear and time-dependent characteristics of electricity demand. This integrative strategy proved superior to stand-alone models, aligning with prior research that emphasized the benefits of combining statistical and machine learning approaches for time-series forecasting [2, 3, 12, 13].

The short-term forecasting improvements, particularly the 13.9% reduction in MAPE relative to the best-performing single model (LSTM), underscore the Hybrid model's adaptability in responding to short-lived fluctuations in electricity demand. These findings reinforce earlier studies where hybridization mitigated the limitations of individual models and improved responsiveness to sudden demand shifts influenced by weather and consumer behavior [1, 4, 6, 10]. Moreover, statistical significance testing confirmed that the Hybrid consistently outperformed LSTM, with both parametric and non-parametric tests indicating robust improvements, consistent with theoretical expectations of ensemble learning reducing variance and enhancing stability [5, 9, 11, 15].

In the day-ahead horizon, where forecast uncertainty generally increases due to accumulated external influences, the Hybrid model retained a clear edge. Its 11.8% improvement in MAPE over LSTM demonstrates its scalability and resilience under volatile market conditions. This result is particularly significant in deregulated electricity markets, where accurate day-ahead load predictions are crucial for unit commitment, bidding strategies, and minimizing imbalance costs [7, 11, 14, 16]. Previous studies have highlighted that even small percentage improvements in forecast accuracy can yield

substantial economic benefits for both utilities and consumers, supporting the operational relevance of hybrid frameworks^[8, 14].

The residual analysis further confirmed the Hybrid's reliability, with error distributions centered around zero and reduced variance compared to single models. This indicates minimal systematic bias, a critical feature for integration into decision-making processes within electricity markets. Such findings echo the outcomes of benchmarking exercises and forecasting competitions, where hybrid or ensemble-based models consistently outperformed single algorithms in robustness and generalizability^[1, 4, 14].

Overall, this research validates the hypothesis that hybridization significantly enhances the predictive accuracy and stability of load forecasting models in electricity markets. By demonstrating consistent improvements across multiple horizons, the study contributes empirical evidence to the growing consensus that hybrid machine learning models represent a viable pathway toward meeting the increasing forecasting demands imposed by renewable integration, demand variability, and market deregulation^[5, 9, 12, 15].

Conclusion

The present study confirms that a hybrid model integrating ARIMA, SVM, and LSTM provides superior performance in electricity load forecasting across both short-term and day-ahead horizons, compared to individual statistical or machine learning models. By effectively capturing the linear, nonlinear, and temporal dependencies of demand data, the hybrid approach significantly reduced forecasting errors and demonstrated robust generalizability under fluctuating market and weather conditions. The results clearly validate the hypothesis that ensemble-based methods enhance stability, minimize variance, and offer a more reliable solution to the inherent challenges of electricity market forecasting. These improvements are not only statistically significant but also carry strong operational implications for power system operators, market participants, and policy makers. Accurate forecasting enhances bidding strategies, reduces imbalance penalties, improves unit commitment scheduling, and supports demand response planning, thereby contributing to the economic efficiency and reliability of electricity markets. Based on these findings, several practical recommendations emerge. First, power utilities and grid operators should prioritize the adoption of hybrid forecasting models as part of their energy management systems, as the demonstrated accuracy gains can lead to substantial cost savings and enhanced reliability. Second, electricity markets should invest in building computational infrastructure that can support the real-time deployment of such models, ensuring their scalability for large datasets and fast-changing conditions. Third, hybrid approaches should be integrated with renewable energy forecasting systems to improve the predictability of variable sources like wind and solar, as this would strengthen the resilience of the grid in high-renewable penetration scenarios. Fourth, regulators and policy makers should encourage utilities to adopt hybrid analytics through regulatory incentives, training programs, and industry standards, thereby promoting innovation and harmonization across the energy sector. Fifth, continuous model refinement should be pursued through periodic retraining using updated datasets, incorporating additional exogenous variables such as social activity patterns,

economic indicators, and extreme weather conditions to further improve predictive power. Finally, collaboration between academia, industry, and governments should be strengthened to accelerate the development of next-generation hybrid models that leverage explainable AI techniques, ensuring transparency, trust, and interpretability in decision-making. Collectively, these measures will enable the energy sector to harness the benefits of advanced hybrid forecasting systems, enhancing efficiency, market stability, and sustainability in the evolving landscape of modern electricity markets.

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